



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(24CI3201)

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TITLE OF THE PROJECT	Agile-Driven DevOps Hotel Management and Reservation Platform	
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Abstract

The Agile-Driven DevOps Hotel Management and Reservation Platform is a web-based application developed to provide an efficient and centralized solution for managing essential hotel operations. Traditional hotel management processes often involve manual handling of room availability, guest information, reservations, check-in, and check-out, which can lead to errors, delays, and difficulty in maintaining updated information. The proposed platform addresses these challenges by integrating major hotel management activities into a single digital system.

The system is designed to manage room details, guest records, room availability, reservations, check-in, check-out, and related hotel operations in an organized manner. The application uses a web-based architecture with Spring Boot for backend development, HTML5, CSS3, JavaScript, and Bootstrap 5 for the user interface, and MySQL for database management. Maven is used for project and dependency management, while Git and GitHub are used for source-code management and team collaboration. Postman is used for testing and validating REST APIs.

The project follows Agile software development principles, allowing the system to be developed incrementally through user stories, product backlog management, task estimation, sprint planning, development, testing, and continuous improvement. This approach helps the team divide the project into manageable modules and incorporate changes based on feedback. DevOps practices are integrated to improve collaboration between development and testing activities and to support continuous integration and continuous delivery (CI/CD).

The platform also provides scope for modern DevOps practices such as Docker-based containerization, automated CI/CD pipelines, and cloud deployment. These practices can improve application portability, deployment consistency, scalability, and maintainability. Security and software quality can also be incorporated into the development lifecycle through code-quality analysis, vulnerability testing, and secure development practices.

Overall, the proposed system aims to provide a reliable, scalable, maintainable, and user-friendly hotel management solution while demonstrating the practical application of Agile methodology, DevOps practices, web technologies, database management, API development, testing, and deployment automation. The project serves as a practical example of how modern software engineering practices can be combined to develop and continuously improve a real-world hotel management and reservation platform.